

MADHUSREE RAVICHANDRAN

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SUMMARY

Motivated physics graduate with strong foundational knowledge in astrophysics and cosmology, seeking an integrated PhD position in physical cosmology. Interested in contributing to research involving supernova datasets, quasar standard candles, and cosmological models. Eager to apply statistical methods and work with large-scale survey data to deepen understanding of dark energy and universe expansion.

SKILLS

- **Data Analysis & Machine Learning:** Linear Regression, Decision Tree Regression, Support Vector Regression, Cross-validation, RMSE
 - **Tools & Development:** HEAsoft(basics), Google Colab, Streamlit, FastAPI, API development for astrophysical data
 - **Programming:** Python (NumPy, SciPy, Matplotlib), Matlab(basics)
 - **Language Test Score:** English (IELTS 6.5)
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EDUCATION

Bachelor of Science in Physics

Institution: Kongunadu Arts and Science college

Completed in 2023
awaiting degree

- **Relevant coursework in Electronics and Robotics:** Designed and developed a Bluetooth-controlled obstacle-avoiding robot, applying physics principles and electronic components for autonomous navigation and control.
 - **Paper presentation:** Presented project at the National Conference on Emerging Trends in Physical Science and Mathematics (ETPSM - 2023) at Dr. N.G.P. Arts and Science College.
 - **CGPA :** 8.5
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RESEARCH EXPERIENCE

Project: Measuring Cosmological Parameters Using Type Ia Supernovae

Institution: ISA Summer Camp, Indian Space Academy

May 2025 - Jun 2025

- **Objective:** Estimate cosmological parameters and investigate the expansion history of the universe using Type Ia supernovae.
- **Methodology:** Performed a non-linear least-squares fit of observed distance moduli vs. redshifts under the flat Λ CDM cosmology model to estimate the Hubble constant (H_0) and matter density parameter (Ω_m).
- **Result:** Estimated $H_0 = 70.05$ km/s/Mpc and $\Omega_m = 0.301$; derived the universe's age as 13.65 Gyr. The observed variation in H_0 between low- z and high- z samples supports the known "Hubble tension."
- **Impact:** Demonstrated how Type Ia supernovae serve as standard candles to trace cosmic expansion and study dark energy.

Project: Photometric Classification of Astronomical Sources ([link](#))

Apr 2025 - May 2025

Institution: Independent research project

- **Objective:** Improve the accuracy of source classification in photometric datasets to support large-scale structure and galaxy evolution studies.
- **Initial Analysis:** Used color-color diagrams for preliminary classification of point and extended sources, but found significant overlap between classes, limiting accuracy.
- **Machine Learning model development:** Applied supervised ML algorithms—K-Nearest Neighbors, Logistic Regression, and Random Forest—for source classification. Random Forest yielded the best performance, effectively handling feature overlap.
- **Result:** Machine learning methods performed better than the conventional methods

Sep 2024 - Oct 2024

Research Internship: Prediction of Dark Matter Halo Concentration using Machine Learning.

Institution: Spartificial.

- **Objective:** Compared ML model performance to make trained models accessible for future dark matter research.
- **Machine Learning Model Development:** Developed and evaluated ML models—including Linear Regression, Decision tree and Support vector - to predict dark matter halo concentration, using the Bolshoi simulation dataset.
- **Web Application Creation:** Created a user-friendly app (Streamlit frontend, FastAPI backend) enabling dataset uploads, predictions, and performance visualizations, with options to download results and convert data to CSV.

RELEVANT CERTIFICATIONS AND TRAINING

A Workshop on Astrophysics

Institution: Indian Institute of Science (IISc), Bengaluru

Aug 2025 (2 days)

- Covered astrophysics fundamentals and current research trends.

Summer School: Python for Physics

Institution: University of California, Berkeley

Aug 2025

- Learned advanced Python tools for physics and data science.

Astrophysics Training Program

Institution: Mechanica with IIT Madras

Feb 2025 - Mar 2025

- Completed comprehensive training in observational astronomy, stellar and galactic astrophysics, cosmology, and astrobiology
- Developed skills in telescope techniques, photometric data analysis, scientific writing, and presentation
- Conducted a **literature review on the Dark Energy Spectroscopic Instrument (DESI)**, exploring its imaging surveys, data pipeline, and role in mapping the large-scale structure of the universe.[\(link\)](#)

Astronomical Data Science Training Program

Mar 2024 - Jun 2024

Institution: Spartificial

- Covered Python programming basics and advanced image processing techniques for astronomical applications.
- Gained hands-on experience in multispectral deep space image analysis, neural networks, and computer vision for astronomy.
- Built a computer vision project for analyzing astronomical data, generating insights from complex celestial images using machine learning. [View Project Repository](#)

SPACE PROBE: The Astronomer's Toolkit

Nov 2023 - Dec 2023

Institution: Naxxatra

- Intensive school on Observational Astrophysics, covering essential techniques and methods for observing celestial objects in X-ray, optical, and radio astronomy, as well as cosmology.
- Gained practical experience with telescopes, spectroscopy, imaging, and data analysis for astronomical research.
- Completed a live **group presentation on solar physics** on YouTube, demonstrating observational astronomy skills. [Watch Presentation](#)

Certificate Programme in Astrophysics

Nov 2022

Institution: Kongunadu Arts and Science college

- Conducted a literature survey on dark matter and dark energy, exploring the structure and expansion of the universe.
- Studied large-scale cosmological models and the evolution of the universe.

PROFESSIONAL EXPERIENCE

Executive Assistant - Technical Support

Edutech Company: Rapid Future Technology

July 2023 - Present

- Led workshops on robotics and coding, teaching high school students practical applications of STEAM concepts with educational robots.
- Assisted in developing Teaching Sky, an educational product focused on astronomy and astrophysics, contributing to the design and curriculum for student engagement in space sciences.

AWARDS AND ACHIEVEMENTS

- **International Asteroid Search Campaign** – Participated in asteroid detection and tracking.
- **Dynamic Physics Forum** – Active member involved in discussions on modern physics and cosmology topics.

REFERENCES

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